Livestock Management in Organic and Biological Agriculture

Land Sciences

May, 2026

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Livestock Management in Organic and Biological Agriculture (A32363)

Short Title:	Organic Livestock Management		
Faculty	Science and Computing		
Department	Land Sciences		
Cluster	Agriculture		
Author	JGERAGHTY		
Credits	10	**Level:**	Postgraduate

Description of Module / Aims

The module will assess the importance of different livestock production systems for human protein needs. Animal nutrition requirements will be appraised with a focus on maintaining condition and health for optimum output. Conversion to organic livestock production will be described and discussed. Pasture production systems including set-stocking, rotational grazing, mob-grazing, and holistic planned grazing will be compared and contrasted with a focus on regenerative agriculture practices. Integrated pest management approaches to prevent and manage common livestock pests and diseases will be considered. The challenge of anti-microbial resistance will be an important aspect of the material covered. The primary objective of the module is for graduates to understand the inter-relationship between achieving optimal nutrition for sustainable livestock production thereby guaranteeing high standards of animal welfare, output and performance.

Programmes

SCIE-0082	MSc in Organic and Biological Agriculture (WD_SOBIA_R)	1 / 2 / M
SCIE-0082	Postgraduate Programmes in Science (WD_SGRAD_XX)	5 / 0 / E

Indicative Content

- Organic and biological livestock production systems, climate impact, resilience, and human protein needs
- Rumen nutrition for balanced dietary intake and optimal meat and milk production
- Monogastric pig and poultry nutrition and dietary requirements for optimal field performance and production
- Dealing with livestock management transition challenges organic, and biological production systems
- Pasture and forage production and management systems to reduce external inputs and associated costs
- Integrated pest management and low-stress handling to maintain animal health and welfare
- One welfare initiative and sustainable anti-microbial management and use in regenerative agriculture systems
- Policy and regulation related to organic livestock production in the F2F Strategy, the CAP, and the Programme for Climate Action

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Demonstrate a detailed knowledge of organic and biological principles for livestock management.
2. Explain and illustrate the impact of poor nutrition on animal performance for sustainable meat and milk production.
3. Apply nutritional needs and requirements to formulate dietary ingredients for the different livestock categories and groups.
4. Compare, contrast and discriminate between different pasture and forage management systems for livestock production.
5. Review different IPM methods and practices to design and develop appropriate treatments for livestock diseases and pests.
6. Judge the importance of, and critically assess and evaluate, the sustainability of anti-microbial use in livestock production.
7. Appraise and justify the impact of livestock production systems on biodiversity, climate, environment, and water quality based on contemporary knowledge and research findings.

Learning and Teaching Methods

- Online lectures, guest lectures and webinars
- Farm and processing visits (meat and dairy) for class assignment and discussion
- Self-directed case-study on sustainable livestock production

Assessment Methods

	Weighing		Outcomes Assessed
Final Written Examination	50%		1,2,3,4,5,6,7
Continuous Assessment	50%		1,2,3,4,5,7
Case Studies		30%	1,2,3,4,5,7
Practical		20%	1,3,4,5,7

Assessment Criteria

<40%: Very little knowledge and understanding of the subject matter. Unable to carry out critical thinking or practical skills.

40%--59%: The learner will be able to demonstrate a good understanding of sustainable livestock management techniques and production of healthy animals. The student will be able to write and present a topical case study on an aspect of livestock production and welfare.

60%--69%: In addition to the above, the student will demonstrate more detailed knowledge of all the module topics covered, including livestock management developments and market trends. The student will also be able to write and present a good quality case study on livestock production and welfare.

70%--100%: The learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes, including the vital role of animal nutrition in livestock health and diseases. Write and present a related high quality case study on livestock management, production and welfare.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Tutorial		12
Practical		12
Online Delivery		24
Independent Learning		222

Essential Material

- McDonald, P., R.A. Edwards, J.F.D. Greenhalgh, C.A. Morgan, L.A. Sinclair and R.G. Wilkinson. _Animal Nutrition_. 7th ed.. UK: Pearson, 2010.
- Voisin, A. _Grass Productivity - An Introduction to Rational Grazing_. USA: Midwest Journal Press, 2014.

Supplementary Material

- Albrecht, W.A. and Charles Walters. _Soil Fertility and Animal Health_. 2nd ed.. Austin, Texas: Acres USA, 2005.